

## **EUPHRESKO II**

### **Topics of 1<sup>st</sup> round of Research Initiation**

**Status of topics: 14.02.2012**

(Information from EUPHRESKO II - Annual project meeting, Rome, Feb. 14<sup>th</sup> 2012)

***Please get in contact with the Topic Coordinator/Project Coordinator in to get more detailed information on the topic!***

#### **01-CEP**

##### **Current and Emerging Phytophthoras: Research Supporting Risk Assessment and Risk Management**

Topic Coordinator: **Alan Inman** (Defra) ([Alan.Inman@defra.gsi.gov.uk](mailto:Alan.Inman@defra.gsi.gov.uk))

Project Leader: Jeff Peters (Fera)

Research Partners: 12(-14)

BE-EV-ILVO, EE-PMK, IE-DAFF, IT-CRA, NL-PPS, NO-BIOFORSK, PT-INRB, ES-UPV, UK-FR, UK-Fera, UK-SASA, UK-AFBI (to be confirmed: SI, LT)

Funds committed: €467,333 (minimum) over about 2 years

Statuts: Initiated 28 Nov 2011 after all LoCs from Funders; work has started - KO Meeting 1-2 March 2012

##### **Key Objectives:**

Diagnostics: *P. ramorum*; *P. lateralis*; *P. kernoviae*; generic methods (mixture of development and validation/ring testing of methods)

Biology and risk: epidemiology for specific species (*P. ramorum*, *P. lateralis*, including use genotyping approaches) to inform management.

##### **Other points:**

Clustering of some existing *P. ramorum* projects – this is the first time such clustering has done within EUPHRESKO

CEP is therefore coordinating more funds than just the approx. € 500k of additional resource committed

Can make best use of outbreak sites across affected countries.

#### **02-PCN (Potato Cyst Nematode)**

##### **Use of novel molecular methods to understand population diversity and its implications on disease management through the use of resistant potato varieties**

Topic and Project coordinator: Alex Reid (SASA) ([alex.reid@sasa.gsi.gov.uk](mailto:alex.reid@sasa.gsi.gov.uk))

Budget: 107 000 €

Status: Work plan under discussion - KO Meeting March 2012

SASA - Alex Reid

Plant Protection Service - Loes den Nijs

ILVO - Nicole Viaene

National Academy of Agrarian Sciences - Liliya Pylypenko

National Centre for Research and Development - Olesia Witowska

This project will deliver a measurement of population diversity amongst European PCN species and store these results in a database. An assessment of the ability of different populations of PCN to overcome resistance in a number of potato varieties will be carried out.

**03-MELOID****Development and validation of innovative diagnostic tools for detection and identification of *Meloidogyne enterolobii* in support of integrated plant protection strategies**

TC: Gabriele Schachermayr ([gabriele.schachermayr@blw.admin.ch](mailto:gabriele.schachermayr@blw.admin.ch))

**Problem(s):** *M. enterolobii* = aggressive, polyphagous root-knot nematode species

- Reliable detection and identification tools are required to develop appropriate phytosanitary measures and management strategies.

**Aims/ expected benefits:**

- Provision of new detection and identification tools in support of plant inspection services and extension personnel. These tools will be tailored and validated for the suspected roots of entry such as soil and plants
- Provision of expertise and training in support of NPPOs. This will be achieved by ringtests on nematode extraction protocols and molecular detection and identification, including a training workshop

|          | <b>Contact for LoC/<br/>Funding Agencies</b>                                               | <b>LoC<br/>prov.</b> | <b>Funds (€)</b> | <b>Research partners</b>                                                               | <b>Institutes</b> |
|----------|--------------------------------------------------------------------------------------------|----------------------|------------------|----------------------------------------------------------------------------------------|-------------------|
| BE       | Maes Martine                                                                               | Y                    | 20'000.-         | Viaene Nicole                                                                          | ILVO              |
|          | <a href="mailto:martine.maes@ilvo.vlaanderen.be">martine.maes@ilvo.vlaanderen.be</a>       |                      |                  | <a href="mailto:nicole.viaene@ilvo.vlaanderen.be">nicole.viaene@ilvo.vlaanderen.be</a> |                   |
| DE       | Steinmüller Silke                                                                          | Y                    | 5'000.-          | Hallmann Johannes                                                                      | JKI               |
|          | <a href="mailto:silke.steinmueller@jki.bund.de">silke.steinmueller@jki.bund.de</a>         |                      |                  | <a href="mailto:johannes.hallmann@jki.bund.de">johannes.hallmann@jki.bund.de</a>       |                   |
| FR       | Anthoine Géraldine                                                                         | Y                    | ?                | Folcher Laurent                                                                        | LSV               |
|          | <a href="mailto:geraldine.antoine@anses.fr">geraldine.antoine@anses.fr</a>                 |                      |                  | <a href="mailto:Laurent.folcher@anses.fr">Laurent.folcher@anses.fr</a>                 |                   |
| IT       | Roversi Pio Federico                                                                       | Y                    | 30'000.-         | Carletti Beatrice                                                                      | CRA-ABP Firenze   |
|          | <a href="mailto:piofederico.roversi@entecra.it">piofederico.roversi@entecra.it</a>         |                      |                  | <a href="mailto:beatrice.carletti@entecra.it">beatrice.carletti@entecra.it</a>         |                   |
| NL       | v. d. Boogert P.H.J.F.                                                                     | Y                    | 14'000.-         | Den Nijs Loes                                                                          | NCR               |
|          | <a href="mailto:p.h.j.f.van.den.boogert@minlnv.nl">p.h.j.f.van.den.boogert@minlnv.nl</a>   |                      |                  | <a href="mailto:l.j.m.f.den.nijs@minlnv.nl">l.j.m.f.den.nijs@minlnv.nl</a>             |                   |
| TR       | Burcak Alev                                                                                | Y                    | 10'000.-         | Toktay H.                                                                              | PPRI              |
|          | <a href="mailto:aburcak@tagem.gov.tr">aburcak@tagem.gov.tr</a>                             |                      |                  | <a href="mailto:toktay@yahoo.com">toktay@yahoo.com</a>                                 |                   |
| UK       | Inman Alan                                                                                 | Y                    | 56'905.-         | Hockland Sue Hazel                                                                     | FERA              |
|          | <a href="mailto:Alan.Inman@defra.gsi.gov.uk">Alan.Inman@defra.gsi.gov.uk</a>               |                      |                  | <a href="mailto:sue.hockland@fera.gsi.gov.uk">sue.hockland@fera.gsi.gov.uk</a>         |                   |
| CH       | Schachermayr Gabriele<br>topic coordinator                                                 | Y                    | 40'000.-         | Kiewnick Sebastian<br>project coordinator                                              | ACW               |
|          | <a href="mailto:gabriele.schachermayr@blw.admin.ch">gabriele.schachermayr@blw.admin.ch</a> |                      |                  | <a href="mailto:Sebastian.kiwenick@acw.admin.ch">Sebastian.kiwenick@acw.admin.ch</a>   |                   |
| <b>8</b> | <b>total</b>                                                                               |                      | <b>175'905.-</b> |                                                                                        |                   |

**Project information**

- 8 project partners, total budget: € 176'000.-
- funding mechanism: non-competitive project (NC)
- **project coordinator:** Sebastian Kiewnick, Agroscope Changins-Wädenswil Research Station ACW, Switzerland

- documents provided to all partners (i.e. short topic description, instruction for elaboration of NC projects, template LoC, common contract conditions);  
LoC submitted by all partners

- draft work-plan and time schedule elaborated (to be finalized)

Timing:

planned duration: 18 month

start of NC projects planned for beginning of 2012

=> project on track

#### ***04-SENDO***

#### **Diagnostic methods for *Synchytrium endobioticum*, especially for pathotype identification**

Topic Coordinator: Alan Inman (Defra) ([Alan.Inman@defra.gsi.gov.uk](mailto:Alan.Inman@defra.gsi.gov.uk))

Project Leader: Gerard van Leeuwen (NL-PD), Kirstin Flaff (DE-JKI), Jeff Peters, Fera for molecular aspects)

Research Partners: 12(-13)

BE-EV-ILVO, BG-BFSA-CLPQ, DE-JKI, IE-DAFF, LT-MoA, NL-PPS(+PRI), UK-Fera, UK-SASA, UA-IPP-UAAS, RU-FGU-VNIIKR (PO joining; RO to confirm)

Funds committed: €200,486 (minimum) over about 2 years

Statuts: Initiated 28 Nov 2011 after all LoCs from Funders

work has started - KO Meeting February 2012

Key Outputs:

Harmonised method(s) for pathotype identification using differential cultivars.

Harmonised molecular protocols for detecting and identifying *Synchytrium endobioticum* in soil and potato material, including standard winter sporangia & DNA extraction protocols (van Gent *et al*, 2009).

A validated molecular method to distinguish pathotype-1(D1) from other pathotypes

Molecular methods for identifying all pathotypes of *Synchytrium endobioticum*, if pyrosequencing investigations show this is feasible (validated, e.g. by ring testing, if time and resources permit).

Other points and benefits:

Can make best use of pathotype collections from participating laboratories

Will produce increased effectiveness and efficiency of diagnostic methods

Contributes to contingency planning and preparedness for potential future spread of pathotypes

An EPPO priority for ring testing of diagnostic methods

Ring testing could be expanded to other laboratories (e.g. via European Mycological Network) if the research consortium agrees

Subcontracting (by UK-Fera) allows participation of NL-PRI (the developers of the molecular pathotype method) to access their expertise and knowledge

**05-DEP2****Assessment of the risk posed by ornamentals (and tomato seeds) infected by Pospiviroids to tomato crops and evaluation of pospiviroid inactivation procedures and detection protocols for seed testing in tomato**

Topic and project coordinator: Thibaut OLIVIER (CRA-W) ([t.olivier@cra.wallonie.be](mailto:t.olivier@cra.wallonie.be))

Budget: 152 000 €

Scheduled start: April 2012

Partners: 6 countries (Poland ?), 7 funders, 7 research institutions

- Walloon Agricultural Research Centre (CRA-W) Thibaut Olivier
  - Institute for Agricultural and Fisheries Research (ILVO) Kris De Jonghe
  - Italian Ministry of Agricultural (MiPAAF) Alberto Masci
  - Agricultural Research Council (CRA) Francesco Faggioli
- Austrian Agency for Health and Food Safety (AGES) Sabine Grausgruber-Gröger
- Slovenian Ministry of Agriculture Forestry and Food (MAFF) Jana Erjavec
  - Agricultural institute of Slovenia Mojca Virscek Marn
- Estonian Ministry of Agriculture (MARDD) Karme Petrutis and Valentina Gusina
- Ministry of Agriculture of Lithuania (LT- MoA) Vaidevutis Sveikauskas

Status of documents:

- ✓ Short topic description (completed by each partner)
- ✓ Common contract conditions (completed by each partner)
- ✓ Letter of Commitment (sent by each funder)
- Waiting for Polish inputs
- A draft of the research project has been sent to each research partner

**06-PHYLIB****Epidemiology and diagnosis of potato phytoplasmas and Candidatus Liberibacter solanacearum and their contribution to risk management in potato and other crops.**

Topic coordinator: David Kenyon (SASA) ([david.kenyon@sasa.gsi.gov.uk](mailto:david.kenyon@sasa.gsi.gov.uk))

Project coordinator: Colin Jeffries (SASA) & Jennifer Hodgetts (Fera)

Budget: 525 048 €

Statuts: Work has started - KO Meeting early March 2012

SASA - Colin Jeffries

Fera - Jennifer Hodgetts

IVIA - Mariano Cambra (Possible collaboration with ICIA - Canary Islands + Polytechnic University of Valencia-UPV + Centro Regional Diagnóstico/Potato, Salamanca + Plant Print Diagnostics SL, Valencia + CIMA-CSIC, Madrid + Fundecitrus-Brazil).

ILVO - Kris De Jonghe

Plant Protection Central Research Institute - Aynur Karahan

University of Helsinki - Minna Pirhonen

Phytolab - Maria Kolber

Development of a database of potato infecting phytoplasmas and *Ca. Liberibacter* spp. collections world wide  
 Mapping phytoplasma and *Ca. Liberibacter* spp. movement in potato and/or in other hosts  
 Establishing the genetic diversity of populations  
 Publication of validated procedures  
 Collection of available diagnostic protocols  
 Development of new protocols if necessary  
 Ring tests or inter-laboratory comparative testing of methods  
 Information that underpins risk assessment and risk management

### **07-GRAFDEPI**

#### **Epidemiological studies on reservoir hosts and potential vectors of Grapevine flavescence dorée (FD) and validation of different diagnostic procedures for GFD**

Topic Coordinator: Sylvia Bluemel (AGES) ([sbluemel@ages.at](mailto:sbluemel@ages.at))

#### Short description:

The Project EUPHRESKO GRAFDEPI plans to increase the knowledge about Flavescence dorée (FD) disease epidemiology, with special focus on the role of reservoir hosts of the phytoplasma alternative to grapevine and on presence of potential insect vectors beside the reported *Scaphoideus titanus*. Moreover the project has the aim to harmonize the phytoplasma detection techniques within the EU performing inter-laboratory comparison of the most common diagnostic procedures to verify their reliability towards validation parameters. All these approaches provide elements for a more efficient control of the disease.

#### Research topics proposed for this project:

- Generate testable hypothesis and conduct studies on transmission mechanisms and dynamics of FD in particular with respect to alternative hosts and potential vectors of FD (case and outbreak investigations)
- Inter-laboratory trials to compare some diagnostic protocols and calculate their validation parameters.
- Designing new surveillance systems for the control of the disease (e.g. design of sampling plans for testing of propagation material, building on new epidemiological data, identification of new or improved control points and strategies).

| Partner <sup>a</sup> | Mechanism <sup>b</sup> | Total potential Budget <sup>c</sup> |
|----------------------|------------------------|-------------------------------------|
|                      | (RP, VP, NC)           |                                     |
| AT-BMLFUW            | VP/NC                  | € 50000                             |
| AT-AGES              | VP/NC                  | € 10000                             |
| BE CRA-W             | VP/NC                  | € 2000                              |
| BE-EV ILVO           | VP/NC                  | € 15000                             |
| IT-MiPAAF            | VP/NC                  | € 50000                             |
| IT CRA               | VP/NC                  | € 0                                 |
| PT – INRB            | VP/NC                  | € 10000                             |
| CH-FOAG              | VP/NC                  | € 60000                             |
| TR- GDAR             | VP/NC                  | € 20000                             |
|                      |                        |                                     |
| total                |                        | € 217000                            |

Research Providers: 10 (+2)

Austria: AT-AGES, Belgium: CRA-W, BE-ILVO

Italy: CRA-PAV, CRA-ABP, DISTA- University of Bologna, University of Milan

Portugal: PT-INRB, *Serbia: LAP, Slovenia: NIB*

Switzerland: CH-ACW Agroscope, Turkey: TR-PPRS

Status of documents

## PROVIDED:

- ✓ - Topic description
- ✓ - SD\_-\_Tasks\_of\_project\_coordinator\_(PC)\_(comp.)
- ✓ Instructions for elaboration of NC-project
- ✓ T\_-\_Application\_Form\_(full\_proposals)\_(VP)

## DONE (except pure NC researchers):

- ✓ Letters of Commitment
- ✓ COMMON EUPHRESKO CONTRACT CONDITIONS

## IN PROGRESS: project proposal preparation

Project Coordinator:

Graziella Pasquini

CRA-PAV Plant Pathology Research Centre, Rome - Italy

Project start date/duration

Start 2012; 24 months.

COMMENTS: it was very difficult to find a project coordinator; & problems to keep the deadlines

***08-APOPHYT***

**Evaluation of factors determining distribution, impact, detection and characterization of apple proliferation and other fruit tree phytoplasmoses in the European Community**

Topic Coordinator: Silke Steinmüller (JKI) ([silke.steinmoeller@jki.bund.de](mailto:silke.steinmoeller@jki.bund.de))

This project will provide more information on phytoplasmas on fruit trees, specially on pathogen detection in propagation and planting material. In addition a comparison between visual inspection and molecular methods. And the evaluation of disease occurrence of AP will be part of the project. Results are necessary for risk analysis and will be useful for future policy decisions in relation to improving phytosanitary regulations.

Important research aspects:

- ❖ Screening nuclear plantings and nurseries for latent infections and distribution of AP, PD and ESFY in the EC
- ❖ Molecular classification of virulence of the infecting AP phytoplasmas
- ❖ Monitoring disease development of uniformly inoculated trees under various climatic conditions and insect-proof containment
- ❖ Improving detection of the fruit tree phytoplasmas by more efficient and more sensitive diagnostic procedures (LAMP, immuno-capture PCR/LAMP)

| Funding partners | Mechanism <sup>b</sup> | Total Budget <sup>c</sup> |
|------------------|------------------------|---------------------------|
|                  | (RP, VP, NC)           |                           |
| DE-BLE (14)      | NC                     | € 25.000                  |
| DE - JKI (15)    | NC                     | € 1.500                   |
| BE CRA-W (5)     | NC                     | € 5.000                   |
| BE-EV ILVO (6)   | NC                     | € 20.000                  |
| AT-BMLFUW (4)    | VP                     | € 50.000                  |
| AT-AGES (3)      | NC                     | € 10.000                  |
| CH-FOAG          | NC                     | € 80.000                  |
| Norway-Bioforsk  | NC                     | € 12.000                  |
| IT-MiPAFF        | NP                     | € 0,000                   |
| IT- CRA-PAV      | NP                     | € 0,000                   |
| total            |                        | €203.500                  |

Research Providers:

Wilhelm Jelkmann, E. Seemüller (JKI-DE)  
 Stéphan Steyer (CRA-BE)  
 Kris De Jongh (Be-ILVO)  
 Helga Reisenzein (AGES-AT)  
 Markus Bünter (CH-FOAG)  
 Graziella Pasquini (IT-CRA-PAV)  
 Arild Sletten (Norway BIOFORSK)  
 + possibly Poland?

Documents provided:

- Topic description
- Instructions for elaboration of NC-project
- Letters of Commitment
- Common EUPHRESKO contract conditions

PC: Wilhelm Jelkmann / JKI + Helga Reisenzein/ AGES

Project start: 2012

Duration: 24 months

Problems to settle on a project coordinator! Currently most of the coordination work is done by the TC

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**09-PHYTFIRE****Phytosanitary diagnostic, on-site detection and epidemiology tools for fire blight**

Topic Coordinator: Sylvia Bluemel (AGES) ([sbluemel@ages.at](mailto:sbluemel@ages.at))

**Short description:**

This project will develop and apply tools to fill fundamental research gaps in pathogen detection and epidemiology that would improve fire blight phytosanitary and control strategies.

Building on monitoring strategies and technology developed in EUPHRESKO 'ErwinDect', FP7 KBBE 'Q-Detect', and Interreg IV 'Gemeinsam gegen Feuerbrand' this project will provide data on spread and development of *E. amylovora* in context of its environment. Recently published detection procedures will be evaluated and novel methods will be developed. Sampling in the experimental orchard "Kirschgartshausen" provides material from a defined fire blight setting for method control.

Methods for epidemiological monitoring of pathogen populations in flowers, vectors and asymptomatic infections, and for pathogen genotyping will be developed. Monitoring methods will be evaluated for integration with fire blight forecasting models prognosis, for assessing fire blight treatments, and for plant inspection strategies.

**Deliverables include:**

- Innovative methods developed (e.g., SNP genotyping, specific PCR for 'new' *Erwinia* spp.)
- Innovative adaptation of methods for phytosanitary applications (e.g., source-tracking, LAMP-PCR)
- Validation of methods for pathogen detection and monitoring
- Generation of new epidemiological data
- Training in fire blight diagnostics
- Briefing paper to assist PPOs in future optimization of phytosanitary strategies

| Partner <sup>a</sup> (in order of EUPHRESKO II Partner No.) | Mechanism <sup>b</sup> | Total Budget <sup>c</sup> |
|-------------------------------------------------------------|------------------------|---------------------------|
|                                                             | (RP, VP, NC)           |                           |
| CH-FOAG                                                     | VP/NC                  | €100000                   |
| AT-AGES                                                     | VP/NC                  | € 10000                   |
| NL-PPS                                                      | VP/NC                  | € 30000                   |
| PT-INRB                                                     | VP/NC                  | € 20000                   |
| TR-MARA-GDAR                                                | VP/NC                  | € 10000                   |
| DE-BLE                                                      | VP/NC                  | € 25000                   |
| DE-JKI                                                      | VP/NC                  | € 1500                    |
| AT-BMLFUW                                                   | VP/NC                  | € 50000                   |
| ES-INIA                                                     | VP/NC                  | € 133330                  |
| RU-FGU VNIKR                                                | VP/NC                  | € 10000                   |
| UA-IPP UAAS                                                 | VP/NC                  | € 4500                    |
| FR-INRA                                                     | VP/NC                  | € 10000                   |
| LT                                                          | VP/NC                  | € 15000                   |
| Total                                                       |                        | € 419330                  |

**Research Providers: 11 (+4)**

Austria: AT-AGES, AT- TUW; Germany: DE- JKI; Lithuania: ??; Netherlands: NL- PPS; Portugal: PT-INRB; Russia: FGU VNIKR, RU; Spain: ES-IVIA; Switzerland: CH-ACW  
 Agroscope, Turkey: TR-EGE; Ukraine: UkrNDSKR  
 Bulgaria: CLP; France: INRA; Italy: Laimburg; Slovenia: NIB; Estonia



Status of documents

## PROVIDED:

- ✓ - Topic description
- ✓ - SD\_-\_Tasks\_of\_project\_coordinator\_(PC)\_(comp.)
- ✓ Instructions for elaboration of NC-project
- ✓ T\_-\_Application\_Form\_(full\_proposals)\_(VP)

## DONE (except pure NC researchers):

- ✓ Letters of Commitment
- ✓ COMMON EUPHRESCO CONTRACT CONDITIONS

## IN PROGRESS: project proposal preparation

Project Coordinator

Brion Duffy, Agroscope ACW - Wädenswil, Switzerland

Project start date/duration

Start Spring 2012; 24 months

COMMENTS: it was difficult to keep the deadlines in the call procedure

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***10-DROSKII (1)*****Damage potential of *Drosophila suzukii* and development of phytosanitary measures**

Topic Coordinator: Silke Steinmöller (JKI) ([silke.steinmoeller@jki.bund.de](mailto:silke.steinmoeller@jki.bund.de))

The project will reveal additional information of the current situation on *D. suzukii* in Europe. Measures for risk management should support the elimination of new introductions at entry points or help minimising impacts and further spread to reduce the rate of spreading and therefore the yield losses.

Information on the susceptibility of different fruit and grape varieties are needed for the risk assessment and for risk management measures.

Research includes:

- ❖ An analysis of available control measures against *D. suzukii* which can be combined for contingency planning purposes and development of control strategies
- ❖ Refinement of monitoring methods should be refined further
- ❖ Survey on the infestation of grapes and grape varieties in Europe
- ❖ the situation in vine growing areas near infested fruit sites.
- ❖ Susceptibility testing of different fruit varieties
- ❖ Testing the efficacy of different biological control methods and other environmentally-friendly measures against *D. suzuki*

| Funding partners    | Mechanism    | Total Budget |
|---------------------|--------------|--------------|
|                     | (RP, VP, NC) |              |
| DE_JKI              | NP           | € 2,000      |
| DE_BLE              | NP           | € 50,000     |
| IT-MiPAFF / CRA-ABP | NP           | €100,000     |
| AT-AGES             | VP/NC        | € 10,000     |
| CH-FOAG / ACW       | NP           | € 100,000    |
| UK-FERA             | NP           | € 34,143     |
| total               |              | € 296,143    |

Research providers:

Sauro Simoni, Pio Federico Roversi (IT-CRA-ABP)

Peter Baufeld (DE-JKI)

Phil Northing, Howard Bell, Andrew Cuthbertson, Ray Cannon (UK-FERA)

Christa Lethmayer (AT-AGES)

Catherine Baroffio (CH-FOAG)

+ possibly Poland?

Documents provided:

- Topic description
- Instructions for elaboration of NC-project
- Letters of Commitment
- Common EUPHRESKO contract conditions

Project coordinator: Sauro Simoni (CRA-ABP-IT)Project start:2012 (1<sup>st</sup> May?)Duration:

24 months

Pest was much faster than the project planning!